Table 2: Results matrix

Method	Criterion	Overall accuracy (Species)		Family accuracy
		Train	Test	
1. Single Decision Tree	Gini	100.0	95.0	96.25
	Entropy	100.0	95.41	96.80
2a. Bagging	Gini	99.97	96.66	97.50
	Entropy	99.98	96.87	97.50
2b. AdaBoost	Gini	97.52	95.34	96.73
	Entropy	99.35	94.16	95.00
3. Random Forest	Gini	100	97.77	98.40
	Entropy	100	97.71	98.33
4. Hierarchical Decision Tree	Gini	100	93.82	**Router: 95.07**
	Entropy	100	94.65	**Router: 95.62**
5. Gaussian Naive Bayes	—	91.47	90.34	91.94

Table 3: Supplementary: hierarchical failure analysis (test set, %).

Criterion	Router acc.	Oracle acc.	(A) Router right, expert wrong	(B) Router wrong
Gini	95.07	98.26	1.25	4.93
Entropy	95.62	98.33	0.97	4.38